SigmaTau.jl: unified, validated and fast time-and-frequency stability analysis in Julia

Every estimator family with per-τ noise identification, EDF and χ^2 intervals in one call · matches Stable32 and allantools · modified-total kernels 1,600 to 4,200× faster · a new Monte-Carlo-calibrated MHTOTDEV

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One API, every estimator, a χ^2 interval with every point

Allan: ADEV, MDEV, TDEV. Hadamard: HDEV, MHDEV, HTDEV. Total: TOTDEV, MTOTDEV, TTOTDEV, HTOTDEV, MHTOTDEV. Thêo1 / ThêoBR / ThêoH. MTIE, TIE rms, PDEV. Time-resolved DADEV / DHDEV. Real-time ADEV / MDEV / HDEV / MHDEV. N-cornered hat. S_y, S_x, $\mathcal{L}(f)$, calibrated power-law synthesis.

Per τ, in the same call: the deviation, the noise type (lag-1 autocorrelation with B1 / R(n) fallback), the Greenhall–Riley EDF, and the χ^2 bounds at 68.3 % (switchable). Total-family bias corrections are on by default and switchable.

Input: phase or frequency, any τ grid; read, detrend, gap-fill and outlier tools; plain-text round-trip of results.

Real-time and multi-clock: ADEV, MDEV, HDEV and MHDEV accumulate in O(1) per sample per τ from a ring buffer of at most $4 \cdot m_{max} + 1$ samples and equal the batch kernels on the samples seen so far (Dobrogowski–Kasznia 2007, extended to third differences); dynamic σ(t, τ) maps; N-cornered hat for N ≥ 3 clocks.

```
p = read_phase("clock.DAT"; tau0 = 1.0)
r = adev(p)            # τ grid, σ_y, noise type, EDF, CI
suite = stability(p; devs = (:adev, :mdev, :hdev, :mhtotdev))
```

	Stable32	allantools	SigmaTau
Platform	Windows GUI, closed	Python library	Julia library + desktop app
χ^2 intervals with per-τ noise ID	yes	manual	every call
Total-family bias correction	TOTDEV, HTOTDEV	none	all five, switchable
MDEV-class Hadamard (MHDEV, HTDEV, MHTOTDEV)	no	no	yes
Thêo1 / ThêoBR / ThêoH	yes	Thêo1 only	yes
Dynamic (time-resolved) deviation	yes	no	yes
Real-time accumulators	no	ADEV, HDEV, TDEV	ADEV, MDEV, HDEV, MHDEV
N-cornered hat	three clocks	three clocks	N ≥ 3
Multithreaded kernels	n/a	no	yes

Deviation, noise type and interval at every τ

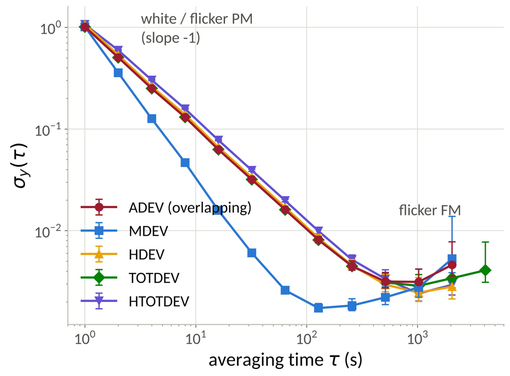


Fig. 1. One call on the shipped 8192-point validation record (τ_0 = 1 s): five estimators with 68.3 % χ^2 intervals from the per-τ identified noise type. The totals reach further (TOTDEV to τ = T/2, HTOTDEV to T/3) with tighter long-τ bounds; MDEV separates white from flicker PM.

Matches Stable32 to its printed precision and allantools to machine precision

Shipped fixture: an 8192-point phase record with Stable32 and allantools outputs beside it. ADEV, MDEV, TDEV and HDEV agree with Stable32 to ≤ 0.005 % (its five significant figures) and with allantools to ≤ 1×10^{-13}; the raw total kernels agree with allantools to ≤ 4.4×10^{-15}; confidence bounds agree with Stable32 to ≤ 0.05 % at 11 of 12 τ. Where a default differs from a reference (MTOTDEV bias policy, noise-ID detrending), the per-τ tables name the keyword that reproduces the reference.

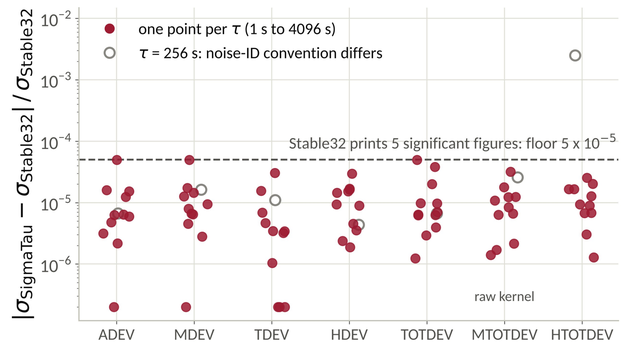


Fig. 2. Relative difference to Stable32 at every τ for seven estimators. Filled points lie at or below the 5×10^{-5} floor set by Stable32's printed precision; open points are the τ = 256 s noise-ID convention (Stable32 α = +2, detrended identifier α = 0). MTOTDEV compared raw, as Stable32 reports the uncorrected statistic.

Modified totals on a 4×10^5-sample record: 0.47 s instead of 30 min

Same samples, same τ grid, same statistic, kernel-only timing; allantools is single-threaded as it has no parallel path. On a measured 406 763-sample phase record MTOTDEV takes 0.469 s against 1 803.6 s and HTOTDEV 0.455 s against 1 928.0 s (3,842× and 4,233×). The gap is algorithmic as well as compiled-versus-interpreted: the reflected extension is streamed through a per-thread buffer and the window sum is split across threads.

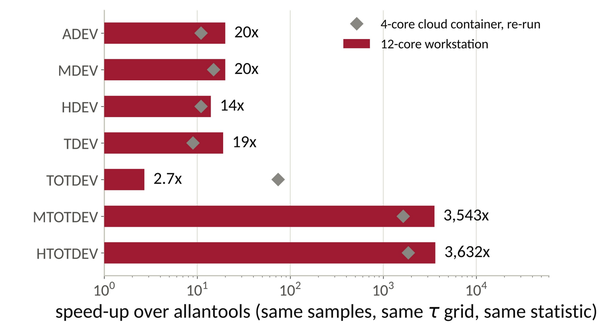


Fig. 3. Speed-up per kernel on an N = 25 000 white-PM record (bench/, 12-core workstation; diamonds: independent 4-core re-run). Ratios are the portable result; absolute times depend on the machine.

MHTOTDEV: a new total estimator, calibrated by Monte Carlo

Modified Hadamard total deviation combines drift rejection (third differences), white-from-flicker PM separation (modified form) and long-τ confidence (reflected extension). No published bias or EDF existed, so both were measured against synthesized noise: α ∈ {+2 … −2}, N ∈ {1024 … 32768}, R = 3000 realizations per cell, one seeded stream per (α, N, r), fits over $\tau/\tau_0 \geq 16$. Bias B = E[MHTOTVAR]/E[MHVAR] against MHDEV on the same realization; EDF from the χ^2 variance relation; bootstrap standard error 2.6 %.

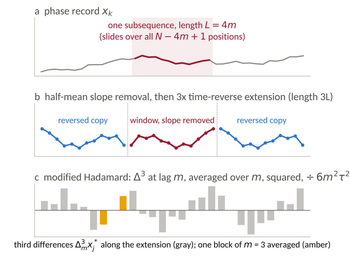


Fig. 4. Kernel (Greenhall scheme): each of the N − 4m + 1 windows of length 4m has its half-mean slope removed and is extended by two time-reversed copies; the third difference at lag m is averaged over m, squared, normalised by $6m^2\tau^2$ and averaged over all windows. The public call removes the least-squares frequency drift first.

Unbiased for PM noise, reads high for red FM; EDF fits with $R^2 \geq 0.9982$

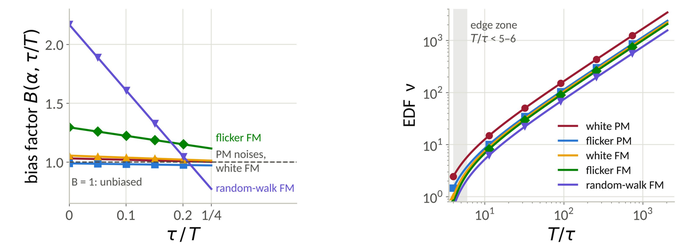


Fig. 5. Shipped models $B = b_0 + b_1 \tau/T$ and $\nu = b\, T/\tau - c$ (seed 20260531). Below T/τ ≈ 5–6 the interval is indicative only; σ itself is valid to the reach limit τ = T/4.

α	noise	bias b_0	bias b_1	EDF b	EDF c	R^2
+2	white PM	1.0308	−0.1197	1.6869	4.3166	0.9982
+1	flicker PM	0.9883	−0.0686	1.1671	3.2107	0.9988
0	white FM	1.0545	−0.1706	1.0861	3.3451	0.9991
−1	flicker FM	1.2947	−0.7193	1.0095	3.3254	0.9991
−2	random-walk FM	2.1686	−5.5884	0.7631	2.4490	0.9985

Reproducible by construction

Shipped with the code: the validation fixture with Stable32 and allantools outputs, the per-τ comparison tables, the benchmark scripts for both libraries, and the Monte Carlo harness with its seed and tracked fit artifact.

Next: real-time monitoring of the laboratory Rb-87 clock comparisons with the Hadamard family; Monte Carlo calibration of the Thêo family and of gapped records.

References

1. W. J. Riley, Handbook of Frequency Stability Analysis, NIST SP 1065 (2008).
2. IEEE Std 1139-2022, Standard Definitions of Physical Quantities for Fundamental Frequency and Time Metrology.
3. C. A. Greenhall and W. J. Riley, Uncertainty of stability variances based on finite differences, Proc. 35th PTTI (2003).
4. D. A. Howe, R. L. Beard, C. A. Greenhall, F. Vernotte and W. J. Riley, A total estimator of the Hadamard function used for GPS operations, Proc. 32nd PTTI (2000).
5. F. Vernotte and D. A. Howe, Generalization of the total variance approach to the different classes of structure functions, Proc. 14th EFTF (2000).
6. A. Dobrogowski and M. Kasznia, Real-time assessment of Allan deviation and time deviation, Proc. IEEE FCS / EFTF (2007).
7. Reference implementations compared: Stable32 (W. J. Riley) and allantools (A. E. Wallin).

Acknowledgments: [funding / program / advisor placeholder]. Documentation: ianlap.github.io/SigmaTau.jl